

8	(a)	(i)	${}^{60}_{27}\text{Co} \rightarrow {}^{60}_{28}\text{Ni} + {}^0_{-1}\text{e} + \gamma$ [B1] for e. [B1] for the other 2.	B1 B1
		(ii)	Number of nuclei in 1 g of C0-60 = $\frac{1}{60} \times 6.02 \times 10^{23}$ $= 1.003 \times 10^{22}$ $A = \lambda N$ $= \frac{\ln 2}{5.272 \times 265 \times 24 \times 60 \times 60} 1.003 \times 10^{22}$ $= 4.182 \times 10^{13} \text{ Bq}$ Correction: 365 days	C1 C1 A1
	(b)	(i)	A discrete packet of energy of electromagnetic radiation (energy > 100 keV OR wavelength < 10 pm.)	B1 B1
		(ii)	When the nucleus de-excites from a higher to lower energy level, a gamma photon of energy equal to the energy transition is emitted. Since photon energy in Fig 8.1 is discrete, the difference in energy levels and hence the energy levels of the nucleus themselves must be discrete.	B1 B1

			(iii) 1. Nucleus Energy Levels / MeV The diagram shows four horizontal energy levels for Ni-60. The y-axis is labeled 'Nucleus Energy Levels / MeV' with values 0, 1.3, 2.2, and 2.5. The levels are labeled E1 (Ni-60) at 0, E2 at 1.3, E3 at 2.2, and E4 (Ni-60) at 2.5. Six transitions are shown with arrows: γ_1 (blue arrow from E4 to E3), γ_2 (blue arrow from E3 to E2), γ_3 (black arrow from E4 to E2), γ_4 (black arrow from E4 to E1), γ_5 (blue arrow from E3 to E1), and γ_6 (black arrow from E2 to E1).	
			Energy of E3 = 2.2 MeV E3 labelled and draw to scale on diagram	B1 B1
			2. The 3 transitions giving rise to γ_1 , γ_2 and γ_5 drawn correctly -1 for any incorrect transition	B2
			3. $\Delta E = \Delta m.c^2$ $(2.5 \times 10^{-6})(1.60 \times 10^{-19}) = \Delta m.(3.00 \times 10^8)^2$ $\Delta m = 4.444 \times 10^{-30} \text{ kg}$ $= 0.0002677 \text{ u}$ Correction: 0.002677 u At excited state, mass = rest mass + Δm = 55.93079+0.0002677 = 59.93347 u	C2

		(ii)	Energy of photon = total kinetic energy lost of beta-particle $\frac{hc}{\lambda_{\min}} = E_{\beta}$ $\frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{\lambda_{\min}} = (0.30 \times 10^6)(1.60 \times 10^{-19})$ $\lambda_{\min} = 4.14 \times 10^{-12} \text{ m}$	C1 A1
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